

PERSONAL INFORMATION

Dr. Nagarajan Ganapathy

 BTBM Building, Department of Biomedical Engineering, IIT Hyderabad, India

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RESEARCH INTERESTS

Artificial intelligence, Machine Learning, Deep Learning, Behavioral informatics, Affective computing, Clinical data science, Computer Vision, Multimodal sensor fusion, Biosignals and image analytics, non-contact vital sign monitoring

Academic Employment

April 2022 - Till	Assistant Professor, Department of Biomedical Engineering, Indian Institute of Technology Hyderabad, Telangana, India
Dec 2020- Mar 2022	Post-doctoral Research Fellow with Prof. Thomas Deserno at PLRI Institute of Medical Informatics, TU Braunschweig, Braunschweig, Germany
Jan 2020- Nov 2020	Senior Research Fellow with Prof. Ramakrishnan Swaminathan at Biomedical Engineering Group, Indian Institute of Technology Madras, India
July 2019- Dec 2019	Institute Pre-Doctoral Fellow with Prof. Ramakrishnan Swaminathan at Biomedical Engineering Group, Indian Institute of Technology Madras, India
Aug 2017– Sept 2018	Visiting Research Scientist at TU Braunschweig, PLRI Institute of Medical Informatics, Braunschweig, Germany

EDUCATION

July, 2019	Doctorate (Ph.D.) in Biomedical Engineering, Indian Institute of Technology (IIT) - Madras, India “Analysis of Electrodermal Activity signals in Arousal and Valence states and its validation with EEG source Localization” under the supervision of Prof. Ramakrishnan Swaminathan at IIT Madras, India (thesis defended in 2019). It won the prestigious Institute pre-doctoral fellowship for finish the thesis in less than four years.
May, 2012	Master of Engineering (M.E.) in Computer Science and Engineering, Anna University, India
April, 2009	Bachelor of Technology (B.Tech) – Information Technology, Anna University, India

PROFESSIONAL RECOGNITION/ACADEMIC AWARDS AND HONORS

2023 – Till	Associate Editor, IEEE Transaction of Affective Computing DAAD Research Ambassador, Indo-German Relationship – 2022-2025 EMBS (125 yr old Society) YP Core Member 2024, 2025 VDBMT – German Biomedical Engineering group AAACI – Affective computing group Associate Editor, IEEE Transaction of Affective Computing
October 2021 – February, 2022	Studienqualitätsmittel (SQM) grant awarded by TU Braunschweig for the topic – “Wearable devices for real-time stress and cognitive workload monitoring.”
July, 2019– January, 2020	Institute Pre-Doctoral Fellowship for finishing the Ph.D. in less than 4 years in IIT Madras
August, 2017– October, 2018	DAAD – Bi-National Guided Research Fellowship Peter L. Reichertz Institute for Medical Informatics, TU Braunschweig, Germany Medical Informatics, Department of Computer Science and Engineering
January, 2015– June, 2019	Half Time Research Assistantship awarded for PhD research at IIT Madras by Ministry of Human Resource and Development, Government of India

RESEARCH CONTRIBUTION STATISTICS:

ORCID ID: 0000-0002-3743-5388; Scopus Author Identifier: 58296234200

Description	Number	Description	Number
Journal Publication	50	Books/Books Chapter	2
Total Citation	859	Patent Published/Filed	13
h-Index	15	Patents Granted	2
I10-index	23	Conference abstract/papers	34
Collaborations	30	Total Funding (in Lakhs)	1248.47

*Citation/h-index/ i10-index Source: Google Scholar

PhD Students Graduated: 2 (Co-PI) OC Completed: 1 Ongoing: 5

CURRENT PROJECTS

Sanctioned:

Sl. No.	As PI or Co-PI	Project Title	Sponsoring Agency	Period of project	Project Cost (in Rs., in Lakhs)
1.	Co-PI	Sankalp: Strengthening Program Implementation and monitoring to achieve single-digit Neonatal mortality	Bill-Melinda Gate Foundation	2022-2027	625.42
2.	PI	SMART-BEING: Smart Multimodal Enabled Affective Computing to Promote Wellbeing	Indo-Japan	2022-2024	20.00
3.	PI	Development of Quantum Neural Networks for Smart Mental Healthcare	MEITY	2022-2024	40.00
4.	PI	Non-Invasive fighter Pilot Health monitoring	Honeywell India	2023-2024	8.00
5.	Co-PI	KeraEyeFATE: Tensor-based Machine Learning for Early Detection of Keratoconus	AI4CIPS	2023-2026	42.08
6.	PI	3D Model Data Generation Solution Using AI-Based Architectural Drawing Technology	Indo-Korean (Cadian)	2024-2026	88.00
7.	PI	MuStPiC: Multimodal Stress Assessment in Fighter Pilots Using Contrastive Learning: A Novel Approach for Real-Time Intervention	Ministry of Education	2024-2026	35.00
8.	PI	SMART-BEING: Smart Multimodal Enabled Affective Computing to Promote Wellbeing	JST-LOTUS	2024	Travel Grant
9.	Co-PI	Towards Multifunctional Two-micron Ultrafast Fiber Laser for High-Precision Biomedical applications	CEFIPRA	2023-2025	84.30
10.	Co-PI	PRATHAME: Partnerships in Research for Advanced Technologies in Healthcare, AI, Materials and Energy	ANRF-PAIR	2025-2027	200.00
11.	PI	Carbon composite-based Flexible Dry Electrodes for non-invasive Long-Term on-skin Biopotential Monitoring	Wadhvani Foundation	2025-27	105.67

Submitted:

Sl. No.	As PI or Co-PI	Project Title	Sponsoring Agency	Period of project	Project Cost (Rs. in Lakhs)	In case project not yet sanctioned, the date of submission of the proposal
1	PI	LLM4MedReg: Design and Development of Large Language Models for Medical Devices Regulation and Standards	ANRF	3	88,81,952	21-04-2025
4	PI	A Multimodal Deep Learning Approach toward Universal Recognition of Emotions while Driving across Cultures	DST-JSPS	3	10,00,000	18-08-2022
5	PI	HOTSPOTS: Context-Aware Salient Regions Prediction in Multimodal data with Epileptic Seizures using Medical Explainable Artificial Intelligence Technologies	Fulbright Fellowship	2	Travel	30-07-2025
6	PI	Development of Quantum Neural Networks for Smart Mental Healthcare	GIAN	2	10,00,00	06-10-2025
7	PI	Development of Multimodal Smart Sport-Related Musculoskeletal Injury Prediction and Rehabilitation Assessment Platform	ICMR	3	86,03,120	Preparation

PUBLICATIONS

A. Publications before joining IITH:

International Journals

- Veeranki, Y.R.; **Ganapathy, N.**; Swaminathan, R. Characterization of dichotomous emotional states using electrodermal activity based geometric features. *Biomed. Sci. Instrum.* 2022, 58, 75–81, doi:10.34107/NSJX733575.
- Shaikh, S.; **Ganapathy, N.**; Swaminathan, R. Automated segmentation of corpus callosum in brain mr images in alzheimer's conditions using improved unet++ model. *Biomed. Sci. Instrum.* 2022, 58, 89–95, doi:10.34107/NSJX733589.
- Warnecke, J.M.; **Ganapathy, N.**; Koch, E.; Dietzel, A.; Flormann, M.; Henze, R.; Deserno, T.M. Printed and Flexible ECG Electrodes Attached to the Steering Wheel for Continuous Health Monitoring during Driving. *Sensors* 2022, 22, 4198. <https://doi.org/10.3390/s22114198>
- Selvaraju, V.; Spicher, N.; Wang, J.; **Ganapathy, N.**; Warnecke, J.M.; Leonhardt, S.; Swaminathan, R.; Deserno, T.M. Continuous Monitoring of Vital Signs Using Cameras: A Systematic Review. *Sensors* 2022, 22, 4097. <https://doi.org/10.3390/s22114097>
- Yedukondala Rao Veeranki, **Ganapathy, N.**, Ramakrishnan S. 2022. "Analysis of Fluctuation Patterns in Emotional States Using Electrodermal Activity Signals and Improved Symbolic Aggregate Approximation", *Fluctuation and Noise Letters*, 21(1), 2250013.
- Y. R. Veeranki, H. Kumar, **N. Ganapathy**, B. Natarajan and R. Swaminathan, "A Systematic Review of Sensing and Differentiating Dichotomous Emotional States Using Audio-Visual Stimuli," in *IEEE Access*, vol. 9, pp. 124434-124451, 2021, doi: 10.1109/ACCESS.2021.3110773
- Tute, E.; **Ganapathy, N.**; Wulff, A. A Data Driven Learning Approach for the Assessment of Data Quality. *BMC Med Inform Decis Mak* 2021, 21, 302, doi:10.1186/s12911-021-01656-x.
- Veeranki, Y.R.; **Ganapathy, N.**; Swaminathan, R. Differentiation of Dichotomous Emotional States in Electrodermal Activity Signals Using Higher-Order Crossing Features and Parametric Classifiers. *Biomed. Sci. Instr* 2021, 57, 322–332.
- Warnecke JM, Wang J, Cakir T, Spicher N, **Ganapathy N**, Deserno TM. Registered report protocol: Developing an artifact index for capacitive

- electrocardiography signals acquired with an armchair. Plos one. 2021 Jul 28;16(7):e0254780.
10. **Ganapathy, N.**, Diana Baumgartel and Thomas M. Deserno. 2021. "Automated Detection of Atrial Fibrillation in ECG signals Using Dynamic Symbol Assignment based Co-occurrence Patterns and Ensemble Learning", *Sensors*, 21(10),3542.
 11. **Ganapathy, N.**; Veeranki, Y.R.; Kumar, H.; Swaminathan, R. Emotion Recognition Using Electrodermal Activity Signals and Multiscale Deep Convolutional Neural Network. *J Med Syst* 2021, 45, 49, doi:10.1007/s10916-020-01676-6.
 12. **Ganapathy, N.**, Ramakrishnan S. and Thomas M. Deserno. 2021. "Adaptive learning and cross training improves R-wave detection in ECG", *Comput Methods Programs Biomed*, 200,105931
 13. **Ganapathy, N.** and Ramakrishnan S. 2020. "Convolution Neural Network based Emotion Recognition using Electrodermal Activity Signals and Time-Frequency Features", *Expert Syst. Appl.*, 159, 113571, <https://doi.org/10.1016/j.eswa.2020.113571>
 14. **Ganapathy, N.**, Ramakrishnan S and Thomas M. Deserno, 2018 " Deep learning on 1D biosignals: a taxonomy- based survey", *Yearb Med Inform*; 27(1):98-109. <https://doi.org/10.1055/s-0038-1667083>.
 15. Kanchana D., **Ganapathy, N.**, and Ramakrishnan S., 2019 "Differentiation of Homogenous and Speckled Objects in HEP-2 Specimen Images using Keypoint Features and an Adaptive Cuckoo Search Algorithm", *IEIE Transactions on Smart Processing and Computing*, 8(1), 38-42.
 16. **Ganapathy, N.**; Swaminathan, R. Emotion Recognition Using Electrodermal Activity Signals and Multiscale Deep Convolution Neural Network. In *ICT for Health Science Research*; IOS Press, 2019; pp. 140–140.
 17. **Ganapathy, N.** and Ramakrishnan S. 2016. "Analysis of stress states using electrodermal activity signals and wavelet transform", *Track O. Image Computing. Biomedical Engineering / Biomedizinische Technik*, 61 (s68).

Conference Proceedings

1. A. Klingenberg, V. Purucker, W. Schüler, N. **Ganapathy, N.** Spicher and T. M. Deserno, "Human activity recognition from textile electrocardiograms," 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), Glasgow, Scotland, United Kingdom, 2022, pp. 3434-3437, doi: 10.1109/EMBC48229.2022.9871210.
2. M. S. Ramakrishnan and N. **Ganapathy, N.** "Phenotypes based Classification of Blood-Brain-Barrier Drugs using Feature Selection Methods and Extreme Gradient Boosting," 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), Glasgow, Scotland, United Kingdom, 2022, pp. 1346-1349, doi: 10.1109/EMBC48229.2022.9871431.
3. H. Kumar, N. **Ganapathy, N.**, S. D. Puthankattil and R. Swaminathan, "Assessment of emotional states in EEG signals using multi-frequency power spectrum and functional connectivity patterns," 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), Glasgow, Scotland, United Kingdom, 2022, pp. 280-283, doi: 10.1109/EMBC48229.2022.9871510.
4. Shaikh, S., **Ganapathy, N.**, & Swaminathan, R. (2022). Automated Segmentation of Lateral Ventricles in Alzheimer's Conditions Using UNET++ Model. *Studies in health technology and informatics*, 295, 511–514. <https://doi.org/10.3233/SHTI220777>
5. Subha Ramakrishnan, M., & **Ganapathy, N.** (2022). Extreme Gradient Boosting Based Improved Classification of Blood-Brain-Barrier Drugs. *Studies in health technology and informatics*, 294, 872–873. <https://doi.org/10.3233/SHTI220612>
6. Veeranki, Y. R., **Ganapathy, N.**, & Swaminathan, R. (2022). Classification of Dichotomous Emotional States Using Electrodermal Activity Signals and Multispectral Analysis. *Studies in health technology and informatics*, 294, 941–942. <https://doi.org/10.3233/SHTI220631>
7. Harshit V., **Ganapathy, N.**, Thomas M. Deserno., 2022, "Video-based Driver Emotional state assessment using hybrid deep spatio-temporal feature learning ", *SPIE Conference*, California, USA. doi: 10.1117/12.2613118
8. Kumar, H., **Ganapathy, N.**, Puthankattil, S. D., & Swaminathan, R. (2022). Classification of Emotional States Using EEG Signals and Wavelet Packet Transform Features. *Studies in health technology and informatics*, 294, 943–944. <https://doi.org/10.3233/SHTI220632>
9. Shaji, S.; **Ganapathy, N.**; Swaminathan, R. Classification of Alzheimer Condition Using MR Brain Images and Inception-Residual Network Model. *Current Directions in Biomedical Engineering* 2021, 7, 763–766, doi:10.1515/cdbme-2021-2195.
10. Rao Veeranki, Yedukondala, **Ganapathy, N.** and Swaminathan, Ramakrishnan. "Electrodermal Activity Based Emotion Recognition using Time-Frequency Methods and Machine Learning Algorithms" *Current Directions in Biomedical Engineering*, vol. 7, no. 2, 2021, pp. 863-866. <https://doi.org/10.1515/cdbme-2021-2220>
11. **Ganapathy, N.**, Jannik Flessner, Frauke Koppelin, Thomas M. Deserno., 2021, "Study Design for Assessing the Impact of Indoor Environmental Quality on Human Physiology", *BMT 2021*, Hannover
12. Sreelakshmi S, **Ganapathy, N.** and Ramakrishnan S. and Thomas M. Deserno., 2021, "Differentiation of Alzheimer conditions in MR Brain Images using inception-resnet network ", *BMT 2021*, Hannover
13. Yedukonadala Rao Veeranki, **Ganapathy, N.** and Ramakrishnan S., 2021, "Electrodermal Activity Based Emotion Recognition using Time- Frequency Methods and Machine Learning Algorithms", *BMT 2021*, Hannover, [Accepted]
14. Himanshu Kumar, **Ganapathy, N.** and Ramakrishnan S., 2021, " EEG based emotion recognition using entropy features and Bayesian optimized random forest ", *BMT 2021*, Hannover, [Accepted]
15. Sreelakshmi S, **Ganapathy, N.** and Ramakrishnan S., 2021, "Differentiation of Alzheimer conditions in MR Brain Images using inception module network", *31st Medical Informatics Europe Conference 2021*, Athens, Greece
16. Shaji, S.; **Ganapathy, N.**; Swaminathan, R. Differentiation of Alzheimer Conditions in MR Brain Images Using a Single Inception Module Network. In *Public Health and Inform.: Proc. of MIE 2021*; IOS Press, 2021; pp. 158–162 ISBN 978-164368185-6.
17. Yedukonadala Rao Veeranki, **Ganapathy, N.** and Ramakrishnan S., 2021, "Non-parametric classifiers based emotion classification using electrodermal activity and modified Hjorth features", *31st Medical Informatics Europe Conference 2021*, Athens, Greece
18. Himanshu Kumar, **Ganapathy, N.** and Ramakrishnan S., 2021, "Emotion recognition in EEG signals using decision fusion based electrode selection ", *31st Medical Informatics Europe Conference 2021*, Athens, Greece
19. **Ganapathy, N.** and Ramakrishnan S., 2020, "Emotion analysis using electrodermal activity signals and spiking deep belief network ", *MIE 2020*, Geneva, Switzerland
20. **Ganapathy, N.** and Ramakrishnan S., 2019, "Differentiation of valence states in localized EEG signals using Motif Patterns ", *4th Russian-German Conference on Multi-Scale Bio-Mathematics*, Lomonosov State university, Moscow Russia
21. **Ganapathy, N.** and Ramakrishnan S., 2019, "Emotion recognition using electrodermal activity signals and multiscale deep convolution neural network ", *Proc. of SPIE Stud Health Technol Inform.* 2019; 258:140.
22. Kumar S, Jaglan D, **Ganapathy, N.**, Thomas M. Deserno, 2019, "A comparison of open source libraries ready for 3D reconstruction of wounds". *Imaging Informatics for Healthcare, Research, and Applications*, SPIE Medical Imaging, Vol. 10954, California, United States.
23. Kanchana D., **Ganapathy, N.** and Ramakrishnan S. (2019) "Binary Grey Wolf Optimizer based Feature Selection for Nucleolar and Centromere Staining Pattern Classification in Indirect Immunofluorescence Images", In *41st Annual IEEE EMBC proc.*, Berlin, Germany
24. Kanchana D., **Ganapathy, N.** and Ramakrishnan S. (2019) "Evaluation of performance of fitness functions in Adaptive Cuckoo search for differentiation of Indirect Immunofluorescence images", In *45th Annual Northeast Bioengineering Conference proc.*, New Brunswick, New Jersey, USA.
25. **Ganapathy, N.** and Ramakrishnan S., 2018, " Stress prediction using electrodermal signals and LSTM recurrent neural networks", *International Brain Mind Institute Symposium*, EPFL, Lausanne
26. **Ganapathy, N.** and Ramakrishnan S, 2017, "Emotion Analysis using Electrodermal Activity signals and Dynamic Restricted Boltzmann Machines", *International conference on Mechanics in Medicine and Biology*, Melbourne, Australia.
27. **Ganapathy, N.** and Ramakrishnan S, 2017, "Emotion Analysis using Electrodermal Activity signals and Spiking Neural Networks", *5th international conference on Computation and Mathematics in Biomedical conference*, Pittsburgh, USA, pp. 1316-17. ISSN: 2227-9385
28. **Ganapathy, N.** and Ramakrishnan S, 2017, "Analysis of emotion using Electrodermal Activity signals and Deep Belief Network", *54th International Rocky Mountain Biomedical symposium*, Colorado, USA

B. Publications after joining IITH with no student from IITH as a co-author: International Journals

- Kumar, H.; **Ganapathy, N.**; Swaminathan, R. Analysis of Dynamics of EEG Signals in Emotional Valence Using Super-Resolution Superlet Transform. *IEEE Sensors Letters* 2025, 9, 1–4, doi:10.1109/LENS.2025.3526907.
- Banik, Sourabh, Kumar, Himanshu, **Ganapathy, Nagarajan** and Swaminathan, Ramakrishnan. "Assessment of EEG-PPG Cross Frequency Coherence under Evoked Emotional Arousal" *Current Directions in Biomedical Engineering*, vol. 10, no. 4, 2024, pp. 49-52. doi: 10.1515/cdbme-2024-2012
- Banik, S.; Kumar, H.; **Ganapathy, N.**; Swaminathan, R. Assessment of Emotion Elicitation Using Multimodal Physiological Sensors and Phase Synchronization. *IEEE Sensors Letters* 2024.
- Kumar, P.S.; Govarthan, P.K.; Gadda, A.A.S.; **Ganapathy, N.**; Ronickom, J.F.A. Deep Learning-Based Automated Emotion Recognition Using Multi Modal Physiological Signals and Time-Frequency Methods. *IEEE Transactions on Instrumentation and Measurement* 2024.
- Govarthan, P.K.; Peddapalli, S.K.; **Ganapathy, N.**; Ronickom, J.F.A. Emotion Classification Using Electrocardiogram and Machine Learning: A Study on the Effect of Windowing Techniques. *Expert Systems with Applications* 2024, 124371.
- S. Banik, H. Kumar, **N. Ganapathy** and R. Swaminathan, "Exploring Central-Peripheral Nervous System Interaction Through Multimodal Biosignals: A Systematic Review," in *IEEE Access*, doi: 10.1109/ACCESS.2024.3394036.
- Remya R. Nair, P.A. Karthick, **Ganapathy, N.**, G. Venugopal, Surface electromyographybased analysis of muscle fiber type variations in the lower limb muscles of sprinters using gray level transformed Morlet scalogram images and Haralick features, *Biomedical Signal Processing and Control*, Volume 93, 2024, 106123, ISSN 1746-8094, <https://doi.org/10.1016/j.bspc.2024.106123>.
- Himanshu Kumar, **Ganapathy, N.**, Subha D Puthankattil, Ramakrishnan Swaminathan, "Analysis of EEG Fluctuation Patterns using Non-Linear Phase Based Functional Connectivity Measures for Emotion Recognition", *Fluctuation and Noise Letters*, <https://doi.org/10.1142/S0219477524500512>
- Y. R. Veeranki, **N. Ganapathy**, R. Swaminathan and H. F. Posada-Quintero, "Comparison of Electrodermal Activity Signal Decomposition Techniques for Emotion Recognition," in *IEEE Access*, vol. 12, pp. 19952-19966, 2024, doi: 10.1109/ACCESS.2024.3361832.
- Aunganesh, K., **N. Ganapathy**, Sivakumaran. N and Karthik, P.A. "Recognition of Lower Limb Movements Using Machine Learning Methods and Bispectral Maps of Wireless sEMG Measurements," in *IEEE Sensors Letters*, vol. 7, no. 9, pp. 1-4, Sept. 2023, Art no. 7004604, doi: 10.1109/LENS.2023.3307108.
- Kumar, B.V.; Swarubini, P.J.; **Ganapathy, N.**; Karthik, P.A.; Ronickom, J.F.A. Evaluating the Effectiveness of Machine Learning in Identifying the Optimal Facial Electromyography Location for Emotion Detection. *Biomedical Signal Processing and Control*, 2023.
- Rostam Niakan Kalhori, S.; Deserno, T.M.; Haghi, M.; **Ganapathy, N.** A Protocol for a Systematic Review of Electronic Early Warning/Track-and-Trigger Systems (EW/TTS) to Predict Clinical Deterioration: Focus on Automated Features, Technologies, and Algorithms. *Plos one* 2023, 18, e0283010.
- Sriram Kumar P, Govarthan PK, **Ganapathy N**, Fredo Ar J. A. Comparative Analysis Of Eda Decomposition Methods For Improved Emotion Recognition. *Journal of Mechanics in Medicine and Biology*. 2023 Apr 8.
- P. Sriram Kumar, P. K. Govarthan, **N. Ganapathy** and J. F. A. Ronickom, "Electrodermal activity-based analysis of emotion recognition using temporal-morphological features and machine learning algorithms", *J. Mech. Med. Biol.*, vol. 23, no. 6, Aug. 2023.

Conference Proceedings

- Banik, S.; Kumar, H.; **Ganapathy, N.**; Swaminathan, R. Assessment of Valance Emotional State Using EEG-EDA Coupling and Explainable Classifiers. *Stud Health Technol Inform* 2024, 316, 953–957, doi:10.3233/SHTI240569.
- P. Kumar Govarthan, P. Sriram Kumar, **N. Ganapathy** and J. F. Agastinose Ronickom, "Investigating Windowing Techniques in Emotion Classification with ECG and Machine Learning," 2023 IEEE 5th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA), Hamburg, Germany, 2023, pp. 348-353, doi: 10.1109/ICCCMLA58983.2023.10346740.
- Barigala, V. K., Sriram Kumar P, Govarthan, P. K., Pj, S., Aasaitambi, M., **Ganapathy, N.**, Pa, K., Kumar, D., & Agastinose Ronickom, J. F. (2023). Identifying the Optimal Location of Facial EMG for Emotion Detection Using Logistic Regression. *Studies in health technology and informatics*, 305, 81–84. <https://doi.org/10.3233/SHTI230429>
- Lebaka, L. N., Sriram Kumar P, Govarthan, P. K., Rani, P., **Ganapathy, N.**, & Agastinose Ronickom, J. F. (2023). Automated Emotion Recognition System Using Blood Volume Pulse and XGBoost Learning. *Studies in health technology and informatics*, 305, 52–55. <https://doi.org/10.3233/SHTI230422>
- V. S. Roha, **N. Ganapathy**, N. Spicher, S. Saha and T. M. Deserno, "Assessment of Driver's Stress using Multimodal Biosignals and Regularized Deep Kernel Learning," 2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), Sydney, Australia, 2023, pp. 1-4, doi: 10.1109/EMBC40787.2023.10340564.
- Sriram Kumar P, Govarthan, P. K., **Ganapathy, N.**, & Agastinose Ronickom, J. F. (2023). Comparative Analysis of Electrodermal Activity Decomposition Methods in Emotion Detection Using Machine Learning. *Studies in health technology and informatics*, 302, 73–77. <https://doi.org/10.3233/SHTI230067>
- Govarthan, P. K., Sriram Kumar P, **Ganapathy, N.**, & Agastinose Ronickom, J. F. (2023). Deep Learning Framework for Categorical Emotional States Assessment Using Electrodermal Activity Signals. *Studies in health technology and informatics*, 305, 40–43. <https://doi.org/10.3233/SHTI230418>

C. Publications after joining IITH with students from IITH as a co-author

International Journals

- Upadhyay S, Haas P, Bollmann J, **Ganapathy N**, Deserno TM. Transmission of Vital Data in Emergencies Using the International Standard Accident Number. *Stud Health Technol Inform*. 2025;332:160-164. doi:10.3233/SHTI251518
- Kaveti Pavan, Ankit Singh, D. S. Pawar and **N. Ganapathy**, "Multimodal Wearable-Based Automated Driver Inattention State Assessment Using Multidevices and Novel Cross-Modal Attention Framework," *IEEE Sensors Letters*, vol. 9, no. 9, pp. 1-4, Sept. 2025, Art no. 6009604, doi: 10.1109/LENS.2025.3596610.
- Mukundan, G.; **Ganapathy, N.**; Badhulika, S. Ni-Fe Layered Double Oxide on Porous Nickel Foam: A Rationalized Approach to Electrochemical Sensing of Atrazine Herbicide in Water Samples. *New J. Chem*. 2023, 47, 20026–20037, doi:10.1039/d3nj03329e.
- Chatur, A.; Haghi, M.; **Ganapathy, N.**; TaheriNejad, N.; Seepold, R.; Madrid, N.M. Advanced Classifiers and Feature Reduction for Accurate Insomnia Detection Using Multimodal Dataset. *IEEE Access* 2024, 12, 150664–150678, doi:10.1109/ACCESS.2024.3456904.
- Mukundan, G., **Ganapathy, N.**, & Badhulika, S. (2023). ZnO nanoparticles-copper metal-organic framework composite on 3D porous nickel foam: a novel electrochemical sensing platform to detect serotonin in blood serum. *Nanotechnology*, 34(40), 10.1088/1361-6528/ace368. <https://doi.org/10.1088/1361-6528/ace368>
- Gayathri Ravichandran, Dinesh Harijan, **Ganapathy, N.**, Ganesan Prabusankar, Abhijit De and Aravind Kumar Rengan, "The Multifaceted Role of Degradable Cobalt Nanoparticles: Dual-Target Starvation and Intracellular Acidification Engendering LC3-Associated Whole-Cell Autophagy", 5 (10), 2726–2738, doi: 10.1021/acsmaterialslett.3c00616
- Malepati L, Hoskere V, **Ganapathy N**, Prakash SS. Segmentation of surface and subsurface damages in concrete structures through fusion of multimodal images using vision transformer. *Automation in Construction*. 2025 Nov 1;179:106469.
- Mohammad Sadik Ali, Monika Pebam, Hima Sree Buddhiraju, Sreenath Dey, Veeresh Bantal, **Ganapathy, N.**, Divyadarshini Madhavan, Aravind Kumar Rengan, "Accelerating Diabetic Wound Healing by Modulating the Inflammatory Environment Using Quercetin-Rosemary Oil Lipid Nanoemulsions with Artificial Intelligence-Based Wound Closure Analysis", *Advanced Therapeutics*, Volume 7, Issue 3, doi: /10.1002/adtp.202300345.
- Swarubini P J, Thomas. M. Deserno and **N. Ganapathy**, An Intelligent Camera-Based Contactless Driver Stress State Monitoring Using Multimodality Fusion. *IEEE Sensors Letters*, vol. 9, no. 9, pp. 1-4, Sept. 2025, doi: 10.1109/LENS.2025.3595917.
- Pavan K, Singh A, Pawar DS, **Ganapathy N**. Assessment of Driver Inattention State Using Multimodal Wearable Signals and Cross-Attention-Driven Hierarchical Fusion. *Stud Health Technol Inform*. 2025;323:121-125. doi:10.3233/SHTI250061
- Swarubini P J, Igasaki T, **Ganapathy N**. Automated Fall Detection in Smart Homes Using Multiple Radars and Machine Learning Classifiers. *Stud*

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Book Chapters:

1. Kavati Pavan Kumar*, Swarubini P J*, **Nagarajan Ganapathy** (2025). Cognitive Artificial Intelligence. In. *Artificial Intelligence and Biological Science*. doi:10.1201/9781003492726-18. *Equal contribution.
2. Kavipriya Murugaiyan*, Swarubini P J*, Kavati Pavan Kumar, Basu Bhattacharjee, Aravind Kumar Rengan, **Nagarajan Ganapathy** (2025). Artificial Intelligence in Digital Biology. In. *Artificial Intelligence and Biological Science*. doi: 10.1201/9781003492726-19. *Equal contribution

Patents:

1. Nagarajan Ganapathy, Kaveti Pavan Kumar, Ramakrishnan Swaminathan, Systems and Methods for Mental Health Assessment Using Differential Self-Attention Mechanisms with Wearables in Individuals with Acute Stress, Indian Patent. (Submitted).
2. Nagarajan Ganapathy, Swarubini P.J, Kaveti Pavan Kumar, A System And Method For Real-Time Contactless Lead-Free Driver Stress Detection In Automotive Environments, Indian Patent. (Submitted)
3. Nagarajan Ganapathy, Hashim P, Swarubini P.J, Kaveti Pavan Kumar, System and Method for Dynamic Region-of-Interest Isolation in Imaging Photoplethysmography for Enhanced Heart Rate Estimation During Driving, Indian Patent. (Submitted)
4. Nagarajan Ganapathy, Podakanti Satyajith Chary, Pithani Teja Venkata Ramana Kumar, System and Method for Interpretable Electrocardiogram (ECG) Analysis using Cross-Attention Enhanced Capsule Networks with Factorized Self-Attention, Indian Patent. (Submitted)
5. Nagarajan Ganapathy, Podakanti Satyajith Chary, Ramakrishnan S, System and method for zero-annotation high-confidence anomaly segmentation with explainable AI-Guided multi-instance prompting, Indian Patent. (Submitted)
6. Nagarajan Ganapathy, Vikash PR, Arunkumar S, Chinmay Kushwaha, Swarubini P.J, Ramakrishnan S, System and Method for Continuous Emotional Health Monitoring Using Camera-based Non-contact Sensing and Instrumentation, Indian Patent. (Submitted)
7. Nagarajan Ganapathy, Shivam Upadhyay, Sruthy Krishna, Lohitaa J, Apparatus for real-time wireless transmission of vital parameters and method of manufacturing the same, Indian Patent, Patent application no. 202441085856, Filed: 08 November 2024, Published: 15 November 2024.
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9. Nagarajan Ganapathy, Manuskandan Ramakrishnan, Satyavrat Govindarajan, Sukanta Kumar Tulo, Swarubini P J, System and Method for Assessment of Ground Glass Opacity in a Computed Tomography Image, Indian Patent, Patent application no. 202441084895, Filed: 6 November, 2024, Published: 15 November, 2024.
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11. Nagarajan Ganapathy, Swarubini P.J, Kaveti Pavan Kumar, Karthick P.A, Nithyanadhan Kanagaraj, An Artificial Intelligence (AI) Driven Non-contact Stress Detection System, Indian Patent, Patent application no. 202441036963, Filed: 10 May, 2024, Published: 17 May, 2024.
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13. Nagarajan Ganapathy, Swarubini PJ, Kaveti Pavan Kumar, System And Method For Non-invasive, Non-contact, And Real-time Vital Sign Monitoring, Indian Patent, Patent application no. 202441037262 , Filed: 11 May, 2024, Published: 17 May, 2024.

Administrative Activities at IITH

1. Hostel Warden – Bhabha Block – 2023 - Till
2. Department International champion for all international affairs.
3. ICCR application committee in-charge
4. Ph.D. Advisor for BME Department
5. Selection Committee for Multiskill Assistant post
6. Selection committee for JT, BME post
7. BME Department PhD Selection Committee
8. M.Tech Co-Ordinator (2023-Till)
9. Faculty Advisor – PhD 2024 Batch
10. Faculty Advisor – M.Tech (2024, 2025 Batch)

Outreach Activities at IITH

1. Seminars/Workshop:
 - a. MMTTP 2025 – Medical informatics and Digital Health 2025
 - b. FORGE 2023, DAAD-based Indo-German Meeting 2023
 - c. nRESIDE 2024 , DAAD-based Indo-German Meeting 2024
2. International Students Exchange/Internship:
 - a. Ameya (2023), University of Konstanz, Germany
 - b. Himanshu Singh (2023), TU Braunschweig, Germany
 - c. Shivam Upadhyay (2024), TU Braunschweig, Germany
 - d. Hashim (2025), Uniklink, RWTH Aachen, Germany
 - e. Paarth (2025), Honeywell Bangalore, India
 - f. Nitila (2025), Phillips Bangalore, India
3. Invited Talks:
 - a. Introduction to Computational Neuroscience 2022, School of Biosciences, IIT BHU, Varanasi, 09 September 2022
 - b. Joint session series GMDS/DGBMT: Risk Prediction in Cardiovascular Diseases, VGBMT, University of Medicine, Gottingen, Germany
 - c. AI-based research on cognitive sciences, 2022, Centre for brain and Cognitive Research, IIT Gandhinagar
 - d. Joint mini-symposium and Workshop - SIAM Conference on Computational Science and Engineering, 2023, Amsterdam
 - e. IITM-ISRO Biomedical Engineering Workshop, June 2025
 - f. Advance traffic safety using ADAS and DMS, workshop, July 2025
 - g. GESUND 2025, IIT Kharagpur, Indo-German Collaboration Workshop, 2025
 - h. IISER Tirupathi, IIT Kharagpur, Indo-German Collaboration Workshop, 2025